

Adroit Technologies

Protocol Driver

Name	Mitsubishi Ethernet 1 st Generation (Q, L, iQ-R and iQ-F)
DLL	MITQJE71



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1. Overview

This driver supports the Mitsubishi 3E Frames protocol, with communication to the Q/QnA/L/R and FX5 Series CPU's.

For further information regarding TCP/IP connectivity refer to Windows Networks Help. The SNMP, WEB and FTP services on the QJ71E71 are not required for Adroit to Mitsubishi communication.

For improved network performance and reliability, it is recommended that you install 2 network adapters into your PC. Use the first adapter for Adroit to Adroit and all other Windows connectivity. The first adapter may have all the default installation bindings. Use the second adapter for Adroit to PLC connectivity and bind only to the TCP/IP protocol for PLC communications.

1.1. Upgrading from original A Series Driver for Q CPU

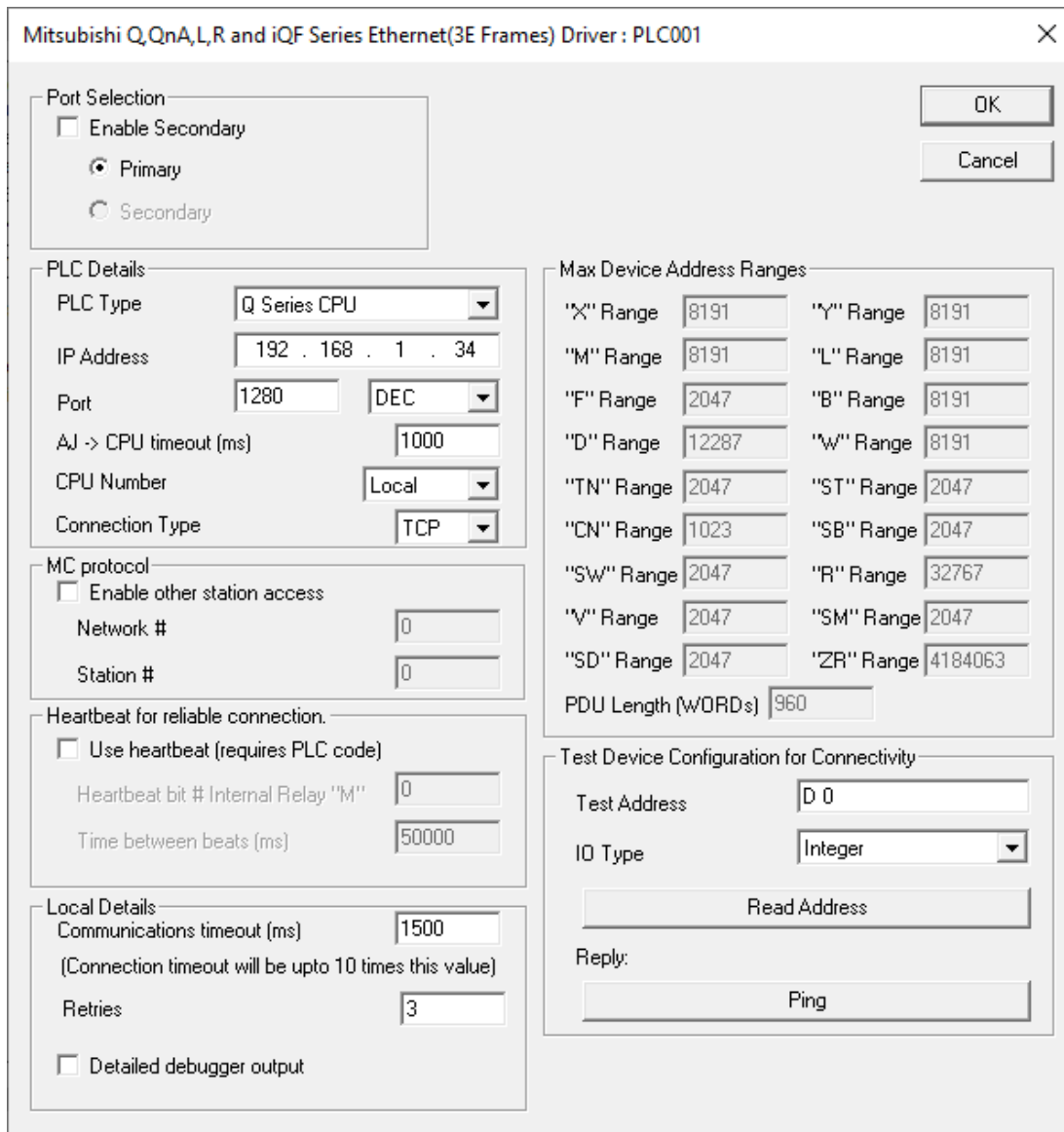
If using the A-series driver, remove all the old Device Agents, and create them again with the same names using the Q driver. Take care to note settings like the IP, port address and watchdog addresses. On the A-Series driver the Decimal or Hex notation used is important and does change over in this Q/QnA driver. Names of I/O types stay the same so far possible, this means that scanned I/O types existing in the A-Series driver that have different names (prefixes) in the Q Driver are promoted automatically to their new names. After saving, the WGP file cannot be loaded on the old driver again without the possibility of losing some scan-points.

2. Wiring

Standard Ethernet.

3. Device Configuration

3.1.1. Device Configuration



The screenshot shows the 'Mitsubishi Q, QnA, L, R and iQF Series Ethernet(3E Frames) Driver : PLC001' configuration window. It is divided into several sections:

- Port Selection:** Includes checkboxes for 'Enable Secondary', 'Primary' (selected), and 'Secondary'.
- PLC Details:** Includes fields for 'PLC Type' (Q Series CPU), 'IP Address' (192 . 168 . 1 . 34), 'Port' (1280), 'AJ -> CPU timeout (ms)' (1000), 'CPU Number' (Local), and 'Connection Type' (TCP).
- MC protocol:** Includes checkboxes for 'Enable other station access', 'Network #' (0), and 'Station #' (0).
- Heartbeat for reliable connection:** Includes checkboxes for 'Use heartbeat (requires PLC code)', 'Heartbeat bit # Internal Relay "M"' (0), and 'Time between beats (ms)' (50000).
- Local Details:** Includes fields for 'Communications timeout (ms)' (1500), '(Connection timeout will be upto 10 times this value)', 'Retries' (3), and a checkbox for 'Detailed debugger output'.
- Max Device Address Ranges:** A table of address ranges for various PLC components.

Range	Value
"X" Range	8191
"Y" Range	8191
"M" Range	8191
"L" Range	8191
"F" Range	2047
"B" Range	8191
"D" Range	12287
"W" Range	8191
"TN" Range	2047
"ST" Range	2047
"CN" Range	1023
"SB" Range	2047
"SW" Range	2047
"R" Range	32767
"V" Range	2047
"SM" Range	2047
"SD" Range	2047
"ZR" Range	4184063
- PDU Length (WORDS):** 960
- Test Device Configuration for Connectivity:** Includes fields for 'Test Address' (D 0), 'IO Type' (Integer), and buttons for 'Read Address', 'Reply:', and 'Ping'.

3.1.2. Port Selection

Enable secondary – Turn this option to have the driver maintain 2 channels of communication, where the radio buttons **Primary** and **Secondary** can then be used to toggle between configurations.

3.1.3. PLC Details

PLC Type – Select the relevant PLC CPU model; Q Series CPU, QnA Series CPU, L Series CPU, R Series CPU, FX5(iQ-F) Series and Custom. By selecting a CPU, you set the device address range. Custom allows you to modify these ranges.

IP Address – Enter the TCP/IP address assigned to the PLC. The default IP address here is your own IP, and often it is required that only the last octet be changed. IPv6 is not supported.

Port – The accessible port on the PLC in GX-developer. This value is expressed in hexadecimal; hence you can use the drop-box next to the entry field to toggle between HEX and decimal display modes.

AJ->CPU timeout (ms) – The timeout setting to be used internally by the external card for transactions with the CPU. This value must be less than the **communications timeout** specified below it. The actual value used is rounded up in increments of 250ms. Legal values are 1 through 63750ms.

CPU Number – Enables communication to up to 4 CPU units on the same PLC. Also allows for selection of “Redundant” CPU pair and if selected will ensure that all communications are transferred to MASTER CPU in configured redundant system.

Connection Type – Select TCP or UDP connection.

3.1.4. MC Protocol

Uses a MELSEC network to access other CPUs remotely through this PLC.

Network No – Network Number of network to be used.

Station No – Remote PLC station number.

3.1.5. Heartbeat for Reliable Connection

Use heartbeat – Enable this button to have the driver keep the PLC connection alive. Please see the PLC program and heartbeat sections below. This option is only useful with the watchdog program running in the PLC.

Heartbeat bit # - Specify the “M” bit number to be turned on by the driver periodically signalling SCADA connection as OK. The default heartbeat bit is “M 0” which must be changed if used.

Time between beats – Specify the millisecond interval at which to turn the bit on. The PLC program should use a time set to double this value to detect link failure.

3.1.6. Local Details

Communications timeout – The interval in milliseconds before the driver assumes the PLC is not responding if no response is received, after which the driver will reset and open the connection. As in the case with a non-fatal error exception response, the driver will re-attempt the transaction 3 times. The number of retries is not settable.

Retries – The number of times a specific transaction will be retried before failing the transaction completely.

Detailed debugger output – turning this option on forces the driver to provide detailed information of activity on the device’s driver diagnostic datascope screen. Use this option when troubleshooting problems involving the watchdog/heartbeat, or specific tags not working.



PLEASE NOTE

The Agent Server must, as always, be restarted for the options changed in this dialog to take effect.

3.1.7. Max Device Address Ranges

This section specifies the maximum address that can be validated for each address type. Note, these ranges can only be edited manually if the PLC Type is set to Custom.

PDU Length (WORDS) – this specifies the maximum number of words (16-bit) that the driver will allow to be transacted in a single transaction.

3.1.8. Test Device for Connectivity

Uses a MELSEC network to access other CPUs remotely through this PLC.

Test Address – Type any valid address in here as found under [Supported Addresses](#).

IO Type – Select the base type of read to be performed (Int/Real/String/Bool).

Read Address Button - Press the button to start a direct connection to the plc from this dialog, it will open a datascope and attempt a read and display the result.



PLEASE NOTE

This must only be executed on the Agent Server machine. If an existing Agent Server connection is currently established, it is recommended to not use this method to determine connectivity. Rather use the existing methods provided in the Agent Server.

Reply – Simple value result is displayed here.

Ping – Click to perform a “Ping” test which will verify access to the PLC card only. Accessibility to the specified port is not verified. For more information in Ping, read the Windows Networking help.

4. Supported Addresses

4.1. Mitsubishi Q series (System Q)

Device	Prefix	Max Range	PLC Data Type	Boolean	Integer	Real	String
Inputs	X	0x1FFF	discrete	X 0			
			Int 16		X0	X0	
			Word 16		X0 W	X0 W	
			Dword 32		X0 D	X0 D	
			Long 32		X0 L	X0 L	
			Float 32		X0 F	X0 F	
Output relay	Y	0x1FFF	discrete	Y 0			
Internal relay	M	8191	discrete	M 0			
Latch relay	L	8191	discrete	L 0			
Annunciator	F	2047	discrete	F 0			
Link relay	B	0x1FFF	discrete	B 0			
Special link relay	SB	0x7FF	discrete	SB 0			
Edge relay	V	2047	discrete	V 0			
Data register	D	12287	bit	D0.0			D 0 Sn
		12287	Int 16		D0	D0	
			Word 16		D0 W	D0 W	
			Dword 32		D0 D	D0 D	
			Long 32		D0 L	D0 L	
			Float 32		D0 F	D0 F	
Link register	W	0x1FFF	Int 16		W 0	W 0	
Timer (current value)	T or TN*	2047	Int 16		TN 0	TN 0	
Retentive timer (current value)	ST	2047	Int 16		ST 0	ST 0	
Counter (current value)	C or CN*	1023	Int 16		CT 0	CN 0	
Special link reg.	SW	0x7FF	Int 16		SW 0	SW 0	
File register	R	32767	bit	R0.0			R 0 Sn
		32767	Int 16		R0	R0	
			Word 16		R0 W	R0 W	
			Dword 32		R0 D	R0 D	
			Long 32		R0 L	R0 L	
			Float 32		R0 F	R0 F	
Extended File register	ZR	4184063	bit	ZR0.0			
		4184063	Int 16		ZR0	ZR0	
			Word 16		ZR0 W	ZR0 W	
			Dword 32		ZR0 D	ZR0 D	
			Long 32		ZR0 L	ZR0 L	
			Float 32		ZR0 F	ZR0 F	

4.2. Mitsubishi QnA series PLC

Device	Prefix	Max Range	PLC Data Type	Boolean	Integer	Real	String
Inputs	X	0x1FFF	discrete	X 0			
			Int 16		X0	X0	
			Word 16		X0 W	X0 W	
			Dword 32		X0 D	X0 D	
			Long 32		X0 L	X0 L	
			Float 32		X0 F	X0 F	
Output relay	Y	0x1FFF	discrete	Y 0			
Internal relay	M	8191	discrete	M 0			
Latch relay	L	8191	discrete	L 0			
Annunciator	F	2047	discrete	F 0			
Link relay	B	0x1FFF	discrete	B 0			
Special link relay	SB	0x7FF	discrete	SB 0			
Edge relay	V	2047	discrete	V 0			
Data register	D	8191	bit	D0.0			D 0 Sn
		8191	Int 16		D0	D0	
			Word 16		D0 W	D0 W	
			Dword 32		D0 D	D0 D	
			Long 32		D0 L	D0 L	
			Float 32		D0 F	D0 F	
Link register	W	0x1FFF	Int 16		W 0	W 0	
Timer (current value)	T or TN*	2047	Int 16		TN 0	TN 0	
Retentive timer (current value)	ST	2047	Int 16		ST 0	ST 0	
Counter (current value)	C or CN*	1023	Int 16		CT 0	CN 0	
Special link reg.	SW	0x7FF	Int 16		SW 0	SW 0	
File register	R	32767	bit	R0.0			R 0 Sn
		32767	Int 16		R0	R0	
			Word 16		R0 W	R0 W	
			Dword 32		R0 D	R0 D	
			Long 32		R0 L	R0 L	
			Float 32		R0 F	R0 F	
Extended File register	ZR	524287	bit	ZR0.0			
		524287	Int 16		ZR0	ZR0	
			Word 16		ZR0 W	ZR0 W	
			Dword 32		ZR0 D	ZR0 D	
			Long 32		ZR0 L	ZR0 L	
			Float 32		ZR0 F	ZR0 F	

4.3. Mitsubishi L series PLC

Device	Prefix	Max Range	PLC Data Type	Boolean	Integer	Real	String
Inputs	X	0x7FF	discrete	X 0			
			Int 16		X0	X0	
			Word 16		X0 W	X0 W	
			Dword 32		X0 D	X0 D	
			Long 32		X0 L	X0 L	
			Float 32		X0 F	X0 F	
Output relay	Y	0x7FF	discrete	Y 0			
Internal relay	M	8191	discrete	M 0			
Latch relay	L	8191	discrete	L 0			
Annunciator	F	2047	discrete	F 0			
Link relay	B	0xFFF	discrete	B 0			
Special link relay	SB	0x7FF	discrete	SB 0			
Edge relay	V	2047	discrete	V 0			
Data register	D	6143	bit	D0.0			D 0 Sn
		6143	Int 16		D0	D0	
			Word 16		D0 W	D0 W	
			Dword 32		D0 D	D0 D	
			Long 32		D0 L	D0 L	
			Float 32		D0 F	D0 F	
Link register	W	0xFFF	Int 16		W 0	W 0	
Timer (current value)	T or TN*	2047	Int 16		TN 0	TN 0	
Retentive timer (current value)	ST	2047	Int 16		ST 0	ST 0	
Counter (current value)	C or CN*	1023	Int 16		CT 0	CN 0	
Special link reg.	SW	0x7FF	Int 16		SW 0	SW 0	
File register	R	32767	bit	R0.0			R 0 Sn
		32767	Int 16		R0	R0	
			Word 16		R0 W	R0 W	
			Dword 32		R0 D	R0 D	
			Long 32		R0 L	R0 L	
			Float 32		R0 F	R0 F	
Extended File register	ZR	524287	bit	ZR0.0			
		524287	Int 16		ZR0	ZR0	
			Word 16		ZR0 W	ZR0 W	
			Dword 32		ZR0 D	ZR0 D	
			Long 32		ZR0 L	ZR0 L	
			Float 32		ZR0 F	ZR0 F	

4.4. Mitsubishi iQ-R Series

Device	Prefix	Max Range	PLC Data Type	Boolean	Integer	Real	String
Inputs	X	0x2FFF	discrete	X 0			
			Int 16		X0	X0	
			Word 16		X0 W	X0 W	
			Dword 32		X0 D	X0 D	
			Long 32		X0 L	X0 L	
			Float 32		X0 F	X0 F	
Output relay	Y	0x2FFF	discrete	Y 0			
Internal relay	M	12287	discrete	M 0			
Latch relay	L	8191	discrete	L 0			
Annunciator	F	2047	discrete	F 0			
Link relay	B	0x1FFF	discrete	B 0			
Special link relay	SB	0x7FF	discrete	SB 0			
Edge relay	V	2047	discrete	V 0			
Data register	D	18431	bit	D0.0			D 0 Sn
		18431	Int 16		D0	D0	
			Word 16		D0 W	D0 W	
			Dword 32		D0 D	D0 D	
			Long 32		D0 L	D0 L	
			Float 32		D0 F	D0 F	
Link register	W	0x1FFF	Int 16		W 0	W 0	
Timer (current value)	T or TN*	1023	Int 16		TN 0	TN 0	
Retentive timer (current value)	ST	0	Int 16		ST 0	ST 0	
Counter (current value)	C or CN*	511	Int 16		CT 0	CN 0	
Special link reg.	SW	0x7FF	Int 16		SW 0	SW 0	
File register	R	32767	bit	R0.0			R 0 Sn
		32767	Int 16		R0	R0	
			Word 16		R0 W	R0 W	
			Dword 32		R0 D	R0 D	
			Long 32		R0 L	R0 L	
			Float 32		R0 F	R0 F	
Extended File register	ZR	4184063	Bit	ZR0.0			
		4184063	Int 16		ZR0	ZR0	
			Word 16		ZR0 W	ZR0 W	
			Dword 32		ZR0 D	ZR0 D	
			Long 32		ZR0 L	ZR0 L	
			Float 32		ZR0 F	ZR0 F	

4.5. Mitsubishi **FX5** series PLC

Device	Prefix	Max Range	PLC Data Type	Boolean	Integer	Real	String
Inputs	X	337	OCT	discrete	X 0		
				Int 16		X0	X0
				Word 16		X0 W	X0 W
				Dword 32		X0 D	X0 D
				Long 32		X0 L	X0 L
				Float 32		X0 F	X0 F
Output relay	Y	337	OCT	discrete	Y 0		
Internal relay	M	8255	DEC	discrete	M 0		
Latch relay	L	0	DEC	discrete	L 0		
Annunciator	F	0	DEC	discrete	F 0		
Link relay	B	0	HEX	discrete	B 0		
Special link relay	SB	0	HEX	discrete	SB 0		
Edge relay	V	0	DEC	discrete	V 0		
Data register	D	8255	DEC	bit	D0.0		
		8255	DEC	Int 16		D0	D0
				Word 16		D0 W	D0 W
				Dword 32		D0 D	D0 D
				Long 32		D0 L	D0 L
				Float 32		D0 F	D0 F
Link register	W	0	OCT	Int 16		W 0	W 0
Timer (current value)	T or TN*	255	DEC	Int 16		TN 0	TN 0
Retentive timer (current value)	ST	0	DEC	Int 16		ST 0	ST 0
Counter (current value)	C or CN*	255	DEC	Int 16		CT 0	CN 0
Special link reg.	SW	0	OCT	Int 16		SW 0	SW 0
File register	R	0	DEC	Bit	R0.0		
		0	DEC	Int 16		R0	R0
				Word 16		R0 W	R0 W
				Dword 32		R0 D	R0 D
				Long 32		R0 L	R0 L
				Float 32		R0 F	R0 F
Extended File register	ZR	0	DEC	Bit	ZR0.0		
		0	DEC	Int 16		ZR0	ZR0
				Word 16		ZR0 W	ZR0 W
				Dword 32		ZR0 D	ZR0 D
				Long 32		ZR0 L	ZR0 L
				Float 32		ZR0 F	ZR0 F

If your PLC's memory range is smaller than the range supported by the driver, it is possible to configure the driver to scan non-existent I/O addresses. Since this can cause unpredictable results, please ensure that all scanned I/O are valid for your controller's current configuration.



INFORMATION

Timer/Counter current values are supported, but the contact and coil values are not scanned. T and C prefixes are promoted to TN and CN respectively.

Adroit string slots and ASCII type is only supported to Data/File registers and can be scanned as "D/R 0 Sn" where n is the number of ASCII characters in the string required, which can go up to a maximum of 79.

Omitted

The following I/O types or devices are not supported:

Device	Mnemonic
Function in/outputs	
Special relays	SM, SD
Timer and counter coils and contacts	TS, TC, SS, SC, CS, CC
Step relay	S
Direct in/out	DX, DY

Non-bit device writes

Bit-writes to items that are not supported in the protocol as discrete (bit devices) will be supported using *read-mask-write*, hence they will be slower.

Bit devices:

Device	Prefix	Base
Input relay	X	HEX
Output relay	Y	HEX
Internal relay	M	Decimal
Latch relay	L	Decimal
Annunciator	F	Decimal
Link relay	B	HEX
Special link relay	SB	HEX
Edge Relay	V	Decimal

1. Bit devices are always accessed in groups of 16 or more, except when writing to a bit.
2. When addressing Integer and Real slots, it is possible to address 32-bit devices in a grouping. A suffix to the scan-address must also be supplied to indicate the size and presentation.
3. Supported suffixes for Integer/Real Adroit slots:

Unsigned 16-bit	W
Signed 16-bit (default)	I
Unsigned 32-bit	D
Signed 32-bit	L
Unsigned 64-bit float	F

Word devices:

Device	Suffix	Base	Boolean	Integer	Real
Data register	D	Decimal	D0.0	D0	D0
Link register	W	HEX	W0.0	W0	W0
Timer (current value)	T or TN*	Decimal	none	TN0	TN0
Retentive timer (current value)	SN	Decimal	none	SN0	SN0
Counter (current value)	C or CN*	Decimal	none	CN0	CN0
Special link reg.	SW	HEX	SW0.0	SW0	SW0
File register	R	Decimal	R0.0	R0	R0
Extended File register	ZR	Decimal	ZR0.0	ZR0	ZR0

1. When accessing a word device in a Boolean or digital tag, a bit number from 0 through 15 must be supplied.
2. Only when accessing a word device in a Real or Analog tag, does the floating-point presentation become available (suffix F).
3. Timers and counter types do not support float "F" presentation.
4. Extended file registers are not supported.



INFORMATION

Timer/Counter current values are supported, but the contact and coil values are not scanned. T and C prefixes are promoted to TN and CN respectively.

5. Protocol Definition and Considerations

5.1. PLC Side Communication setup

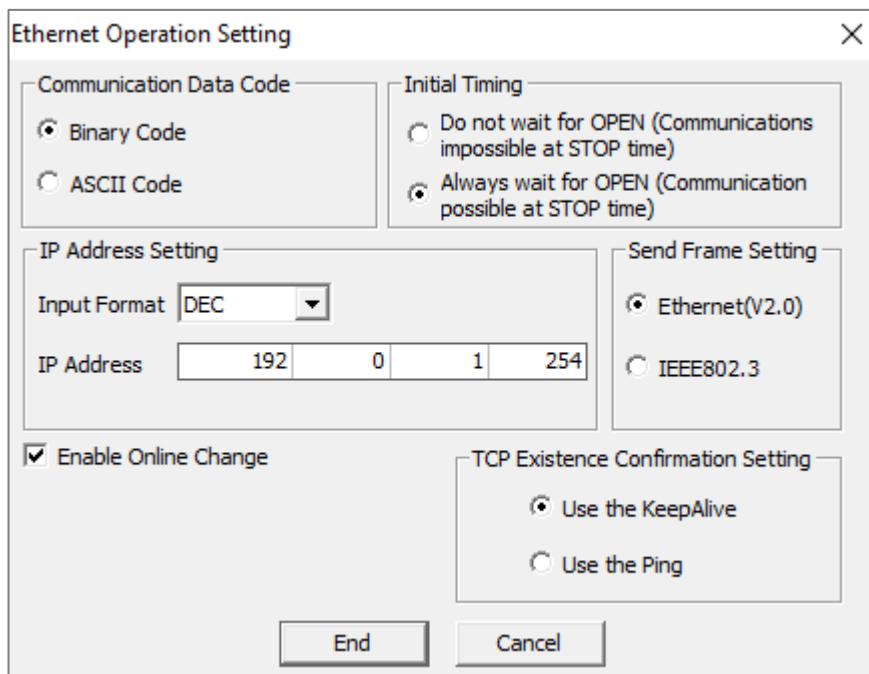
2 possible configurations are presented, one without watchdogs, and one with a watchdog. The driver thus supports 2 modes, with heartbeat, and without respectively. PLC logic is required for the watchdog. If a watchdog/heartbeat is not used, the AJ must simply be configured using GX developer. Sample code using IEC to configure the AJ is not supplied. Any sample programs are supplied as-is and are not under any warranty.

5.2. Using GX Works2 (Used for All except **R Series** CPU)

5.2.1. Settings for CPU without an on-board Ethernet Connection (Q02)

Operational settings

In the Ethernet Network Parameters of the PLC Program specify the Ethernet module in the required slot and then select "Operation Setting". Modify settings based on the image below with the Ethernet modules IP Address in it.



The dialog box titled "Ethernet Operation Setting" contains the following settings:

- Communication Data Code:** ☒ Binary Code, ☐ ASCII Code
- Initial Timing:** ☐ Do not wait for OPEN (Communications impossible at STOP time), ☒ Always wait for OPEN (Communication possible at STOP time)
- IP Address Setting:** Input Format: DEC, IP Address: 192, 0, 1, 254
- Send Frame Setting:** ☒ Ethernet(V2.0), ☐ IEEE802.3
- ☒ Enable Online Change
- TCP Existence Confirmation Setting:** ☒ Use the KeepAlive, ☐ Use the Ping

Buttons: End, Cancel

Open Settings

Configure the connections as below (note how the port address 1280 is in decimal format) Copy the connection as many times as needed for additional ports.

IP Address/Port No. Input Format									
	Protocol	Open System	Fixed Buffer	Fixed Buffer Communication	Pairing Open	Existence Confirmation	Host Station Port No.	Destination IP Address	Destination Port No.
1	TCP	Unpassive	Send	Procedure Exist	Disable	Confirm	1280		

Save, and download the program (you may need to cycle power on the PLC).

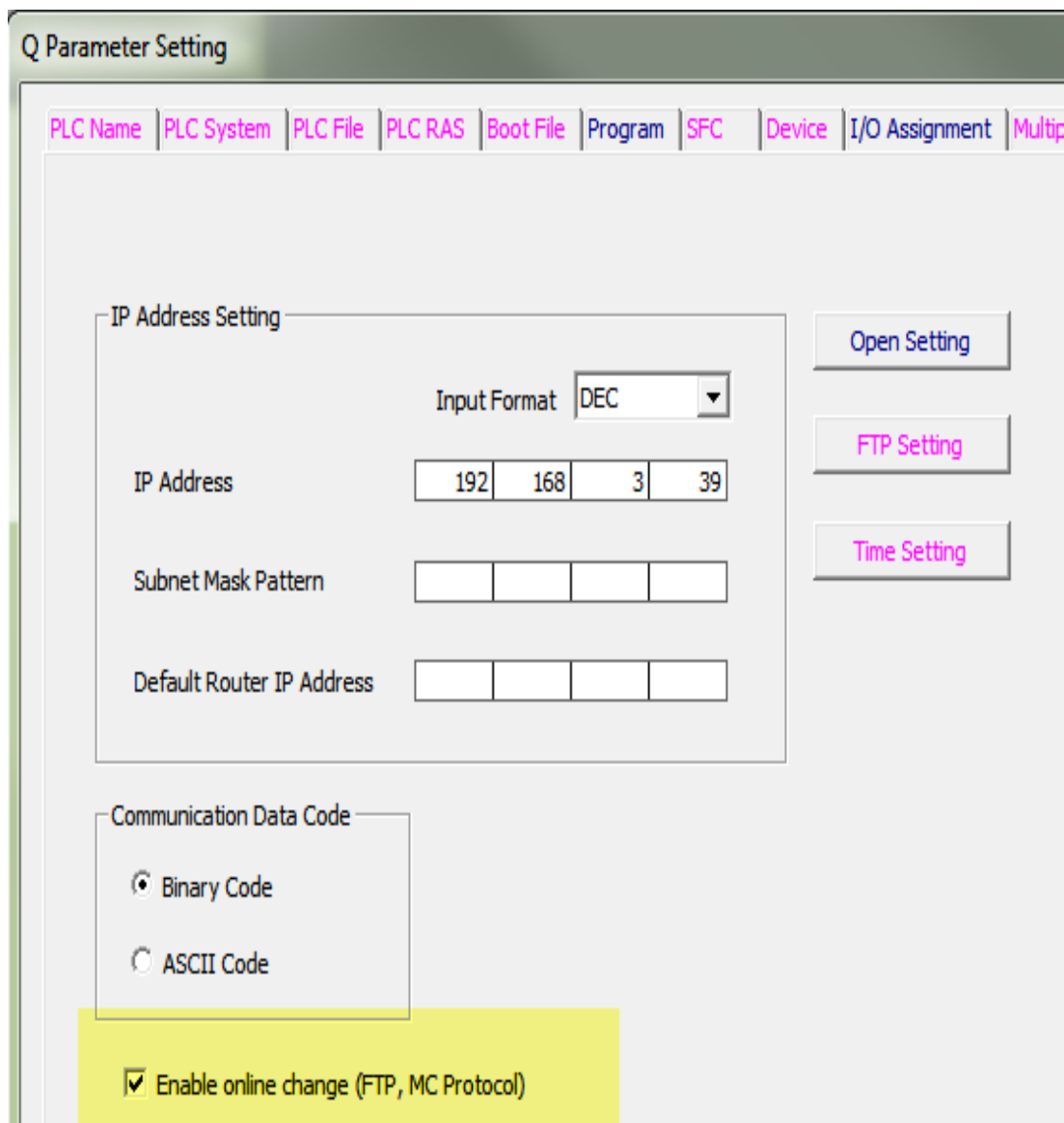
Initial Settings

The default initial settings work very well, however to more easily recover broken network connections, the confirmation existence option above can be changed to "confirm", and the set of Initial settings used in the watchdog section may also be used. Using "Confirm" connections uses a TCP/IP keep-alive, which will cause the PLC to detect and re-open a port in some of the possible network failure conditions.

5.2.2. Settings for CPU with an on-board Ethernet Connection (Q03UDE)

Operational settings

In the PLC Parameters settings of the PLC Program is a TAB called "Build-in Ethernet Port Setting". Select this TAB and specify the Ethernet module settings based on the image below with the Ethernet modules IP Address in it.



Q Parameter Setting

PLC Name | PLC System | PLC File | PLC RAS | Boot File | Program | SFC | Device | I/O Assignment | Multiple

IP Address Setting

Input Format: DEC

IP Address: 192 | 168 | 3 | 39

Subnet Mask Pattern: | | | |

Default Router IP Address: | | | |

Communication Data Code

☒ Binary Code

☐ ASCII Code

☒ Enable online change (FTP, MC Protocol)

Open Setting

FTP Setting

Time Setting

Open Settings

Configure the connections as below (note how the port address 0500 is in hexadecimal format for 1280) Copy the connection as many times as needed for additional ports.

×
Built-in Ethernet Port Open Setting

IP Address/Port No. Input Format DEC

	Protocol	Open System	TCP Connection	Host Station Port No.	Destination IP Address	Destination Port No.
1	UDP	MELSOFT Connection				
2	TCP	MELSOFT Connection				
3	TCP	MELSOFT Connection				
4	TCP	MELSOFT Connection				
5	TCP	MELSOFT Connection				
6	TCP	MELSOFT Connection				
7	TCP	MELSOFT Connection				
8	TCP	MELSOFT Connection				
9	TCP	MELSOFT Connection				
10	TCP	MELSOFT Connection				
11	TCP	MELSOFT Connection				
12	TCP	MELSOFT Connection				
13	TCP	MELSOFT Connection				
14	TCP	MELSOFT Connection				
15	TCP	MELSOFT Connection				
16	TCP	MC Protocol		1280		

(*) IP Address and Port No. will be displayed by the selected format.
 Please enter the value according to the selected number.

End
Cancel

Save, and download the program (you may need to cycle power on the PLC).

Initial Settings

The default initial settings work very well, however, to more easily recover broken network connections, the confirmation existence option above can be changed to "confirm", and the set of Initial settings used in the watchdog section may also be used. Using "Confirm" connections uses a TCP/IP keep-alive, which will cause the PLC to detect and re-open a port in some of the possible network failure conditions.

5.3. PLC setup with PLC-SCADA Watchdog

The watchdog mode allows the driver to recover from the following scenario: unplug network in the middle of a transaction; wait 5 seconds; reconnect. In this scenario, the driver thinks the PLC has gone away, and re-opens the connection, but the PLC does not know the PC is having problems, and thus denies the new connection, since it only allows one connection at a time.



DISCLAIMER

Copies of the Mitsubishi PLC software for the Q(02) is available on a No Guarantee basis and the standard terms of Adroit software distribution apply. Advanced Works 110 will therefore not accept any responsibility for any direct, indirect, or consequential damages of any nature that may occur as a result of the use of such software.

5.3.1. Initial Settings

Timer setting

Module will operate with default values if setting is left blank

	Setting value	Default value	In units
TCP ULP timer	20	60	X500ms
TCP zero window timer	4	20	X500ms
TCP resend timer	4	20	X500ms
TCP end timer	12	40	X500ms
IP assembly timer	3	10	X500ms
Response monitoring timer	30	60	X500ms
Destination existence conformation starting interval	10	1200	X500ms
Destination existence conformation interval timer	10	20	X500ms
Destination existence conformation resend timer	2	3	Times

DNS setting

Input format 

IP address of DNS server 1					
IP address of DNS server 2					
IP address of DNS server 3					
IP address of DNS server 4					

The timeouts have been reduced in order to detect the connection break sooner. Save, and download the program (you may need to cycle power on the PLC).

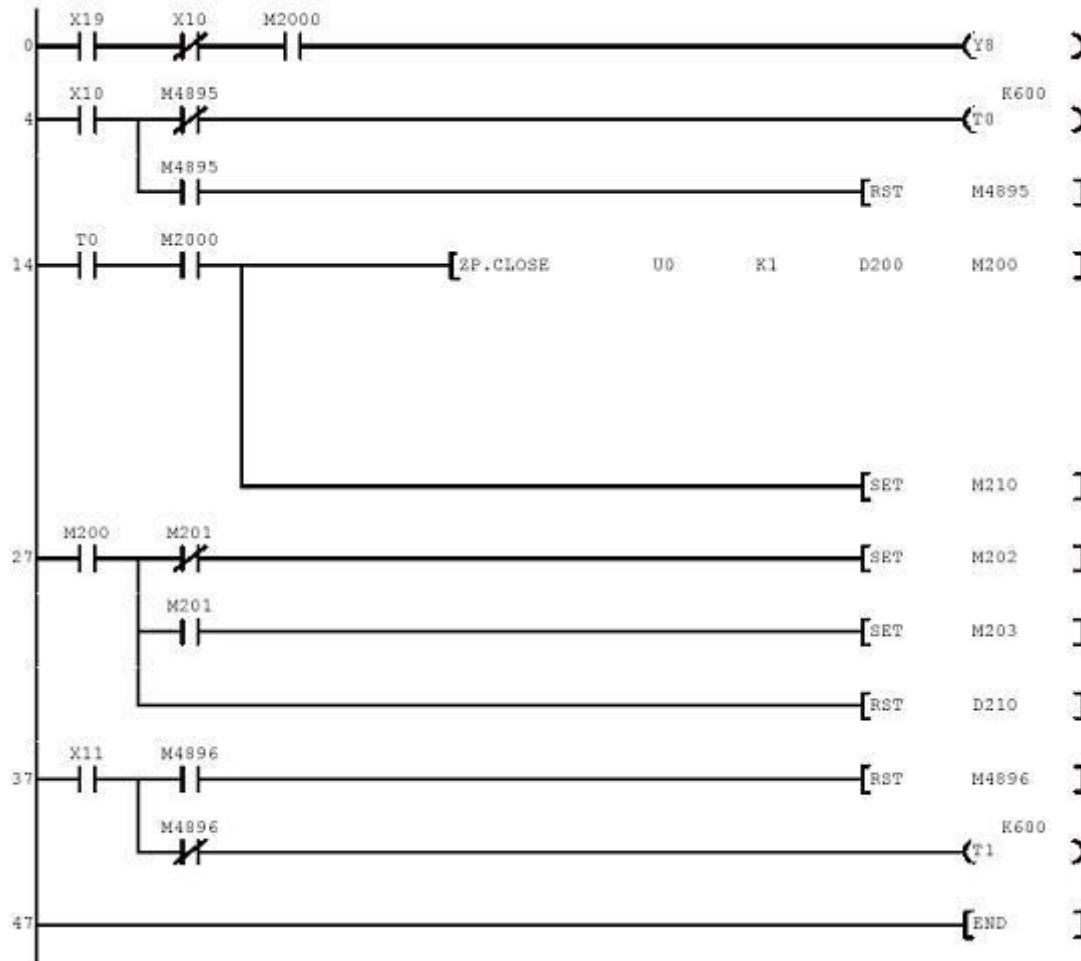
5.3.2. PLC Watchdog Program

Additionally, the watchdog portion of the program used for the A-Series controller may be used to ensure that connections which are terminated unexpectedly due to hardware faults will still be recovered by the AJ on the System-Q. This operates on the basis of the program running a timer for the watchdog bit while off, when the watchdog bit turns on, the timer is reset. If the timer exceeds a

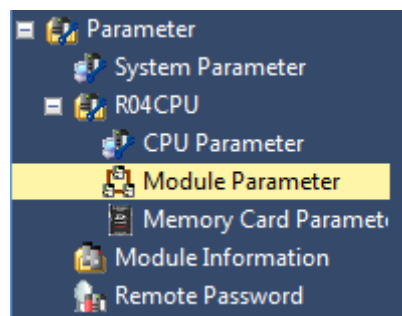
count equal to twice the watchdog interval specified in the Adroit driver, the PLC will shut the connection and allow it to be re-opened by the Adroit driver again.

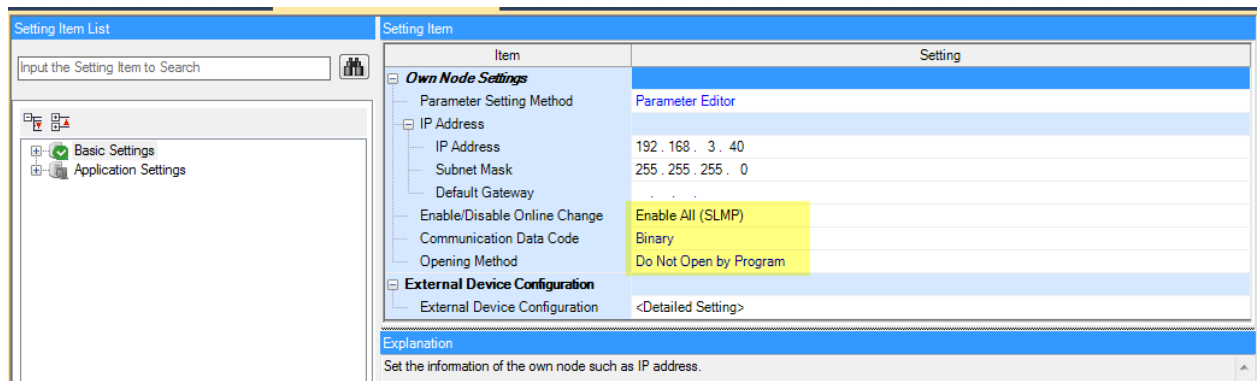
Since the driver keeps turning the bit on periodically, the timer should never expire in normal operation. When it does, the PLC program knows the PC was unable to set the bit, because it cannot or is not talking. One timer and watchdog bit is allocated per PLC TCP/IP port. Please see reference for the AJ for I/O numbers for the consecutive ports.

The PLC Program required if a watchdog is used is shown below. This program shows 2 ports being maintained.



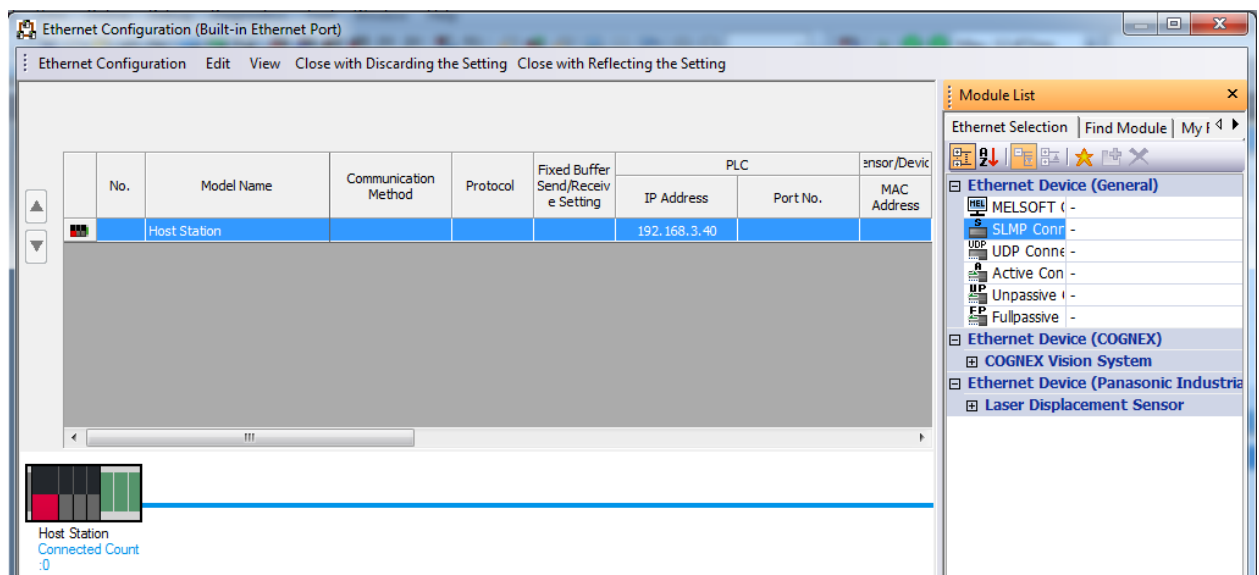
5.4. Use the GX Works3 Software(For R and FX5 Series CPU's)



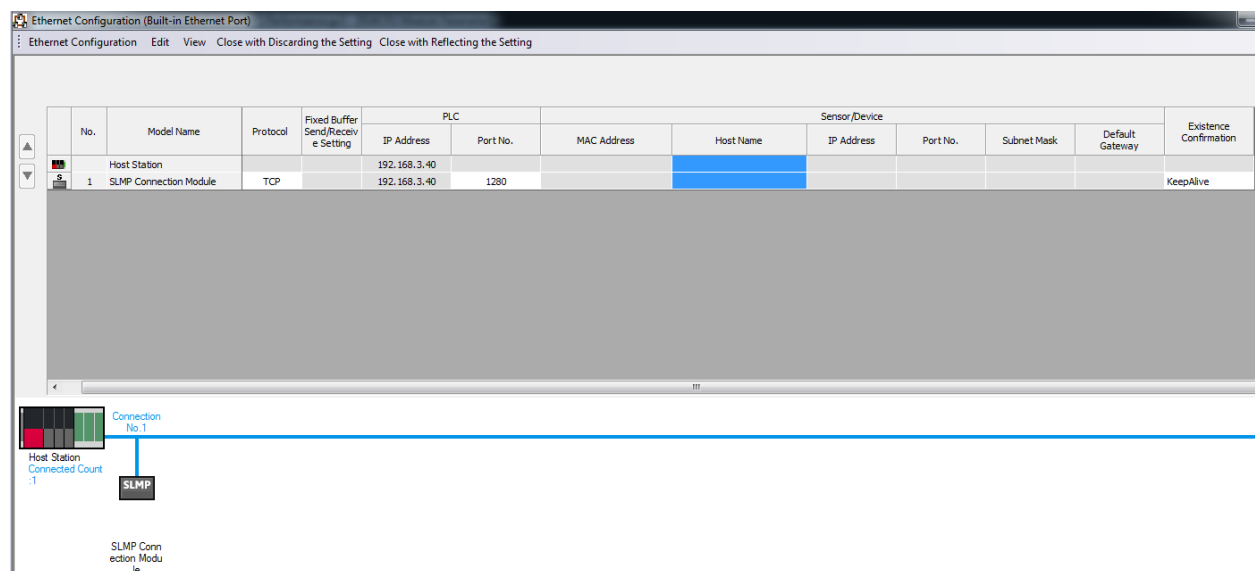


Enter IP address of the module and also do ALL settings marked as yellow in the above Screenshot.

Then select the External Device configuration, Detailed Setting:



Drag and Drop a SLMP connection into the Main window!



Specify the TCP protocol, Port No. 1280 decimal and "keepAlive" existence check. Download those parameters to the PLC and reset the PLC.

5.5. The MC Protocol 3E Frame [Binary]

5.5.1. Request-frame Format

All multi-byte fields are ordered L-H (LSB first) in binary, and MSB first in ASCII frames.

Header	Q Header						Text
	Network no.	PC no.	Dest module I/O no.	Dest module station no.	Request length	CPU Timer	
50 00	00	FF	FF 03	00	20 00	10 00	

Request length is # bytes from this field onwards in the frame. For ASCII frames, this is the total number of characters in subsequent fields. For binary frames, this is the number of bytes.

CPU timer is in 250ms increments 16(dec) = 4000ms.

5.5.2. Response-frame Format

Header	Q Header						Text
	Network no.	PC no.	Dest module I/O no.	Dest module station no.	Request length	Response code	See below
50 00	00	FF	FF 03	00	20 00	00 00	

5.5.3. Text portion of the 3E/3C Frame

Batch Read from word device (read "D2")

Command	Sub-command	Head device number	Device code	# of device points
04 01	00 00	02 00 00	A8	01 00

Batch Read from bit device (read bit "X2")

Command	Sub-command	Head device number	Device code	# of device points
04 01	00 01	02 00 00	9C	01 00

Batch Write to word device

Command	Sub-command	Head device number	Device code	# of device points	Data...
14 01	00 00	00 00 00	A8	01 00	XX XX

Batch Write to bit device (write "Y0")

Command	Sub-command	Head device number	Device code	# of device points	Data...
14 01	00 01	00 00 00	9D	01 00	XX XX

For an ASCII frame, the device-Head and Device-Code (4 bytes) are reversed.

5.5.4. Device Codes

Device	ASCII Frame	BINARY Frame
Special relay ♀	SM	91
Special register ♀	SD	A9
In	X*	9C
Out	Y*	9D
Internal relay	M*	90
Latch relay	L*	92
Annunciator	F*	93
Edge Relay	V*	94
Link relay	B*	A0
Data register	D*	A8
Link register	W*	B4
Timer – contact ♀	TS	C1
Timer – coil ♀	TC	C0
Timer – current ♀	TN	C2
Retentive timer – contact	SS	C7
Retentive timer – coil	SC	C6
Retentive timer – current	SN	C8
Counter – contact ♀	CS	C4
Counter – coil ♀	CC	C3
Counter – current ♀	CN	C5
Special link relay	SB	A1
Special link register	SW	B5
Step relay ♀	S*	98
Direct input ♀	DX	A2
Direct output ♀	DY	A3
Index register ♀	Z*	CC
File Register	R*	AF
	ZR	B0

♀ Types with ♀ are not supported by the driver

* may use space or * in ASCII protocol frame

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

6.3. Troubleshooting Steps

1. The RUN and INIT lamps on the AJ must be lit.
2. The COM.ERR lamp indicates a protocol or hardware fault has occurred.
3. Ping the relevant PLC Ethernet card E.g.
ping 192.168.1.11
The ping should be echoed back.
4. During a ping, the RD and SD lamps on the external module will flicker together.
5. Check that the port is not already open, if another device is already connected to the PLC, the OPEN lamp will be lit. It is possible that multiple pieces of equipment are connected and have an open port. To see the status of open ports, the "Ethernet diagnostics" menu in GX Developer.

6.4. Error Message Reference:

Only commonly encountered PLC error codes are listed, for a full reference, see **MELSEC Q Programmable Controller User manual (basic) QJ71E71**. The PLC error codes can be seen in the Adroit event log, alarm viewer and driver datascope screens, where a reason description also appears.

HEX Code	Remedial Action
0	No error
4031	The scanned device or I/O does not exist. Verify PLC device range configuration or existence of File register.
C050	Configure the Q71 card for binary and restart the module.
C051 - C055	The number of points to read/write is incorrect, contact Adroit technical support.
C056	The scanned device or I/O does not exist. The starting address + length of the request exceeds the number of points in the PLC. Verify PLC device range configuration.
C057	Requested data length miss-match, contact Adroit technical support.

C058	Requested data length miss-match, contact Adroit technical support.
C059	The command issued is invalid, contact Adroit technical support.
C05B	The requested device does not exist. Verify PLC device configuration.
03E8 - 4FFF	Are generated by the CPU, see CPU error code manual for more detail.

In the event of remedial action failing to rectify a problem, the context of the error message is needed by our team. Please see the Adroit help file on how to capture driver data screens by searching for "to create a protocol dump" (no quotes) in the search tab.

6.5. Error Messages

AJ is not in ASCII mode? - Cannot communicate, check AJ mode : Check the mode on the E71 is in ASCII, and then cycle power on CPU.

7. Revisions

7.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1	09-Jan-2023	K. Robinson	Grammar fixes, image alignment changes.	D. Jordaan
2.0				
3.0				

7.2. DLL Revision

Rev.	Date	Changes
82	13-Jul 2022	Very fast scanning could cause high CPU load , fixed.

